

# Rowing



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## Performance Demands of Rowing

This chapter describes the sport of elite rowing, the various factors that are important for success in the standard Olympic distance 2000 m event, and how high-intensity interval training (HIIT) contributes to physiological development.

## Sport Description and Factors of Winning

Rowing, often referred to as crew in the United States, is a sport with origins back to ancient Egyptian times, and it is one of the oldest Olympic sports. The sport is based on propelling a boat (racing shell) on water using oars. By pushing against the water with an oar, force is generated to move the boat. Modern rowing as a competitive sport can be traced to the early 10th century when races were held between professional watermen on the river Thames in London, United Kingdom. Amateur competition began toward the end of the 18th century with the arrival of boat clubs at the British public schools of Eton College and Westminster School, progressing to present day's annual Boat Race held between the University of Oxford and the University of Cambridge crews. The International Rowing Federation

(in French, the Fédération Internationale des Sociétés d'Aviron, or FISA), responsible for international governance of rowing, was founded in 1892 to provide regulation at a time when the sport was gaining popularity. Across six continents, 150 countries now have rowing federations that participate in the sport.

Each year the World Rowing Championships, staged by FISA, sees 22 boat classes raced. In Olympic years only the non-Olympic boat classes are raced at the World Championships. Rowing competitions take place over 2000 m for both men and women. Races are held for sculling (where the rower holds two oars) and sweep boats (where each rower holds one oar), and include pairs, fours, and eights, as well as sometimes the inclusion of a coxswain, although "coxed" fours and pairs are no longer events in the Olympic schedule. The single scull boat is raced by an individual rower (both genders) using two oars. There are additionally lightweight rowing events for both male and female divisions. The different types of rowing classes result in marginal variation in the time taken to complete the 2000 m event, ranging from 5 min (men's eight rowing) to 7 min (women's single scull). The Olympic boats (as of the last Olympics) are outlined in table 16.1. The table shows the associated abbreviations that will be used throughout the chapter.

**Table 16.1** Olympic Rowing Boat Classifications

| Boat                       | Number of athletes in boat | Rowing type | Abbreviation |
|----------------------------|----------------------------|-------------|--------------|
| Men's single scull         | 1                          | Scull       | M1x          |
| Women's single scull       | 1                          | Scull       | W1x          |
| Men's pair                 | 2                          | Sweep       | M2-          |
| Women's pair               | 2                          | Sweep       | W2-          |
| Men's double               | 2                          | Scull       | M2x          |
| Women's double             | 2                          | Scull       | W2x          |
| Men's lightweight double   | 2                          | Scull       | LM2x         |
| Women's lightweight double | 2                          | Scull       | LW2x         |
| Men's four                 | 4                          | Sweep       | M4-          |
| Men's lightweight four     | 4                          | Sweep       | LM4-         |
| Men's quad                 | 4                          | Scull       | M4x          |
| Women's quad               | 4                          | Scull       | W4x          |
| Men's coxed eight          | 8 + 1 × cox                | Sweep       | M8+          |
| Women's coxed eight        | 8 + 1 × cox                | Sweep       | W8+          |

Classifications as of the Rio Olympic Games.

Rowers are typically large in terms of both body mass and stature. This is advantageous for creating a long distance per stroke. Anthropometric values for open-class males and females are  $94.3 \pm 5.9$  kg and  $76.6 \pm 5.2$  kg, respectively. Comparatively, lightweight rowers, who are classified in their division by weight, average  $72.5 \pm 1.8$  kg and  $58.5 \pm 1.5$  kg, respectively, while typical statures average  $1.95 \pm 0.05$  m and  $1.81 \pm 0.05$  m for open-class males/females and  $1.82 \pm 0.04/1.69 \pm 0.05$  m for lightweights (7).

The body composition of rowers is varied, but generally athletes have higher levels of fat-free mass versus other endurance-like sports. This is because their weight is supported by the boat. The average sum of 8 skinfolds in elite-level rowers for males and females has been reported to be  $65.3 \pm 17.3$  mm and  $89.0 \pm 23.6$  mm, respectively. Conversely, lightweight rowers generally carry very little fat-free mass (figure 16.1), possessing an average sum of 8 skinfolds of  $44.8 \pm 8.1$  mm and  $59.7 \pm 12.4$  mm for males and females, respectively, levels near those of other endurance-sport athletes (i.e., triathletes, cyclists, runners) (7).

Like many other water-based sports (i.e., swimming, kayak, etc.), rowing has a large technical component that is needed to apply force effectively in the water medium. Rowing is characterized by a complex sequence of movements, which in its simplest terms can be broken down into the *drive phase* and the *recovery phase*. This is what allows the athlete to propel the boat from one point to the next

in the fastest possible time. Coaches in their training programs nearly always emphasize rowing technique, with specific drills often used to improve distance per stroke and synchronicity/timing between crewmembers. Often, there can be discrepancies between performance on the indoor Concept 2 rowing machine (which requires only raw power and little technique) and that of on-water performance (which is highly technical), thus showing the importance of on-water rowing technique (13). Raw physical power by itself cannot win in this sport.

Traditionally, the intensity of rowing is set by stroke rate (SR) rather than heart rate or speed. The SR chosen during 2000 m racing depends on the boat, with smaller boats rating at lower SRs (30 to 34 strokes per minute; SPM) and larger boats at higher SRs (38 to 41 SPM). The relationship between SR and speed is generally linear (8), making this an important consideration in training program design. Figure 16.2 shows a typical training intensity prescription for an elite M2-, based on the linear speed-to-SR model (8). Interestingly, despite the differences in the SRs associated with race pace, the typical approach I have observed among rowing coaches is one where SR is set at a fixed rate across multiple crew boats, regardless of the chosen SR during competition. This results in a varied training intensity between boats training at the same time. For example, if an M1x has a racing SR of 31, and an M8+ has a racing SR of 39, both might still be asked to rate at 25 SPM during a training session completed together. As a



a



b

**Figure 16.1** The distinctly different physical and proportional anthropometric characteristics of (a) open-class and (b) lightweight rowers.

a. Christian Petersen/Getty Images; b. Joel Ford/Getty Images.

| World record training zones |       |          |           |                           |                  |                    |
|-----------------------------|-------|----------|-----------|---------------------------|------------------|--------------------|
| Grade                       | Boat  |          |           |                           |                  |                    |
| elite                       | M2-   |          |           |                           |                  |                    |
| World                       | Speed | Time per |           |                           |                  |                    |
| (%)                         | km/hr | 500 m    |           | HIIT format               | HIIT targets     | Target stroke rate |
| 77                          | 15.0  | 1:59.6   | 18        | Aerobic endurance (LIT)   |                  | 18                 |
| 78                          | 15.2  | 1:58.1   |           |                           |                  |                    |
| 79                          | 15.4  | 1:56.6   |           |                           |                  |                    |
| 80                          | 15.6  | 1:55.2   | 20        |                           |                  | 20                 |
| 81                          | 15.8  | 1:53.7   |           |                           |                  |                    |
| 82                          | 16.0  | 1:52.3   | 20        |                           |                  | 20                 |
| 83                          | 16.2  | 1:51.0   | 23        |                           |                  | 23                 |
| 84                          | 16.4  | 1:49.7   |           |                           |                  |                    |
| 85                          | 16.6  | 1:48.4   |           |                           |                  |                    |
| 86                          | 16.8  | 1:47.1   |           |                           |                  |                    |
| 87                          | 17.0  | 1:45.9   | 26        |                           |                  | 26                 |
| 88                          | 17.2  | 1:44.7   | 28        | Threshold intensity (MIT) |                  | 28                 |
| 89                          | 17.4  | 1:43.5   |           |                           |                  |                    |
| 90                          | 17.6  | 1:42.4   |           |                           |                  |                    |
| 91                          | 17.8  | 1:41.2   |           |                           |                  |                    |
| 92                          | 18.0  | 1:40.1   | 32        |                           |                  | 32                 |
| 93                          | 18.2  | 1:39.1   | 34        |                           |                  | 34                 |
| 94                          | 18.4  | 1:38.0   |           |                           |                  |                    |
| 95                          | 18.6  | 1:37.0   |           |                           |                  |                    |
| 96                          | 18.8  | 1:36.0   |           |                           |                  |                    |
| 97                          | 19.0  | 1:35.0   | 35        | Long intervals            | Type 3 and 4     | 35                 |
| 98                          | 19.1  | 1:34.0   |           |                           |                  |                    |
| 99                          | 19.3  | 1:33.1   | 37        |                           |                  | 37                 |
| 100                         | 19.5  | 1:32.1   | 38        | Short intervals           | Types 1, 2, 3, 4 | 38                 |
| 101                         | 19.7  | 1:31.2   | 39        |                           |                  | 39                 |
| 102                         | 19.9  | 1:30.3   | 40        |                           |                  | 40                 |
| 103                         | 20.1  | 1:29.4   | 41        | Repeated sprint training  | Type 4 and 5     | 41                 |
| 104                         | 20.3  | 1:28.6   |           |                           |                  |                    |
| 105                         | 20.5  | 1:27.7   |           |                           |                  |                    |
| 106                         | 20.7  | 1:26.9   | Open rate | Repeated sprint training  | Type 5           | Open rate          |
| 107                         | 20.9  | 1:26.1   |           |                           |                  |                    |
| 108                         | 21.1  | 1:25.3   |           |                           |                  |                    |
| 109                         | 21.3  | 1:24.5   |           |                           |                  |                    |
| 110                         | 21.5  | 1:23.7   |           |                           |                  |                    |

**Figure 16.2** Typical training intensity prescription model for an elite M2- based on the general linear speed-to-SR relationship. Recommendations for stroke rate relative to training intensity, HIIT type, and boat speed (percent of world best time) are shown.

result, the intensity will be much higher for the single scull rower compared to the M8+ crew. Figure 16.2 shows how this linear relationship can be extrapolated to training intensity prescription across typically performed training sessions for an M2-.

When comparing across boats, rowing coaches often use a prognostic model, which was developed by British Rowing in the 1980s. This model compares the speed at which a boat travels relative to current world best time (WBT) over 2000 m in that boat class.

## Physical Performance Contributions to Winning

As successful rowing performance depends on correct technique mastery combined with physical capacities, rowers typically perform large volumes of training. Training regimes each week classically consist of around 20 h of training, rowing 150 to 200 km on the water. This volume-based training approach means that the training intensity distribution is generally described by a polarized model, with most training taking place at an intensity below the lactate threshold (LT<sub>1</sub>; figure 16.3; see chapter 3).

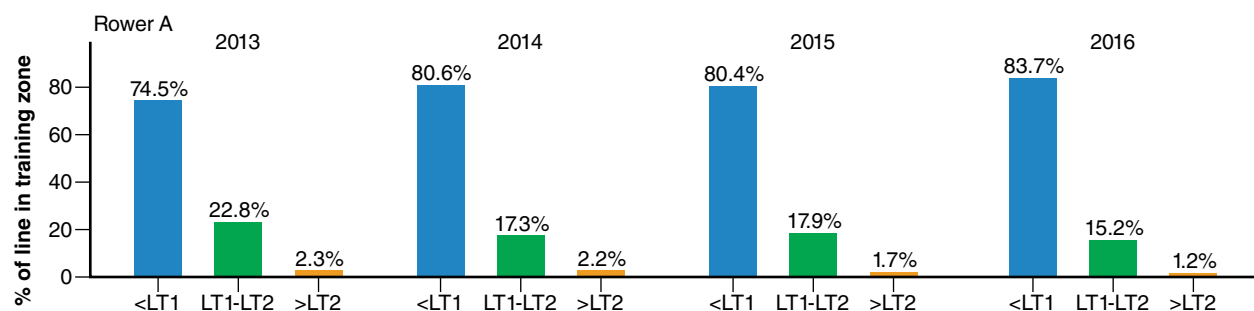
The physiological attributes that determine success in rowing are well documented. The strongest predictor of 2000 m rowing performance is the power output at  $\dot{V}O_2$ max and maximal power/force during 5 peak strokes ( $r=0.95$ ), highlighting that both aerobic fitness and strength play key roles. Other strong relationships include absolute  $\dot{V}O_2$ max ( $r=0.88$ ) and the  $\dot{V}O_2$  at blood lactate threshold or power associated with a 4 mmol<sup>-1</sup> blood lactate reading ( $r=0.92$ ; i.e., LT<sub>2</sub>) (5, 21). As such, it is clear that successful rowing performance is highly dependent on both aerobic and anaerobic energy production, with both the power at  $\dot{V}O_2$ max and 5 stroke peak power showing strong correlations with performance (5).

Specific weight training generally constitutes a significant proportion of a rower's typical training week (although not in every case (16)), where rowers generally complete 2 or 3 weight-training sessions per week. Rowers are relatively strong (i.e., high force produced for specific movement tasks) when compared to other endurance athletes (3), with elite male rowers displaying the ability to generate a force of more than 2000 N in an isometrically simulated rowing position. In layman's terms, this means rowers for 1 RM are able to deadlift, back squat, and bench row 1.9, 1.9, and 1.2 times their body weight (10). Accordingly, strength clearly plays an important role in rowing performance, with studies showing the bench pull power and 1 RM power-clean as strongly related to peak stroke power (11).

## Targets of Physical Performance of Rowing

As mentioned, there are clear contributions from both the aerobic/anaerobic energy systems and strength in rowing. Estimated requirements of both aerobic and anaerobic systems have been estimated at 70% to 80% and 20% to 30%, respectively, over a 2000 m ergometer performance (12). While the aerobic pathway may be the dominant system used across a 5 to 7 min all-out event, both systems must be addressed appropriately in training so as to ensure event success (figure 16.4). As pacing strategies normally take a "U"-shaped approach during 2000 m races, the anaerobic pathway may be of primary importance during the initial sprint and final phases of the race.

Given that the fastest time over the 2000 m event in the world is 5:18.7 (men's eight rowing), versus 7:07.7 for the women's single scull (slowest Olympic



**Figure 16.3** The yearly training intensity distribution of an Olympic champion rower each year prior to the Olympic Games or World Championships event. LT<sub>1</sub>: first lactate threshold; LT<sub>2</sub>: second lactate threshold.



**Figure 16.4** The position of the rower on the three axes shows that rowing is a sport requiring high levels of strength and endurance.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

**Table 16.2** Global Physiological Targets for a Rower Based on Event and Its Respective World Best Times

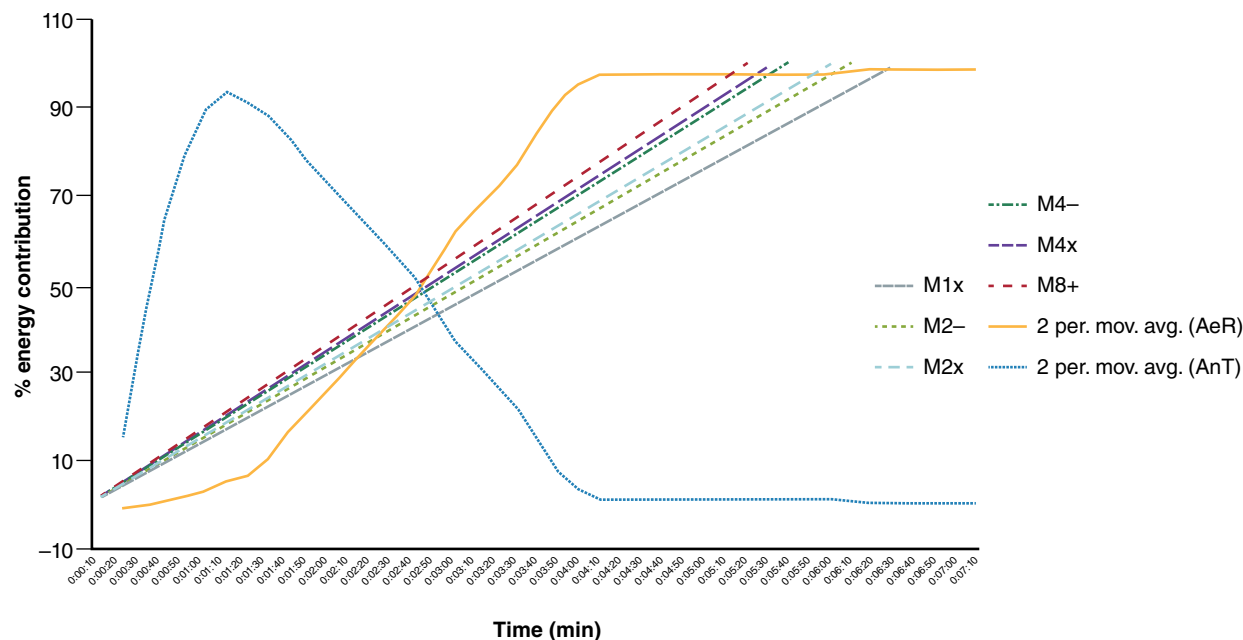
| Event                  | $\dot{V}O_2\text{max}$ | Anaerobic capacity | PS  | $VT_1$ | $VT_2$ | Male WBT (August 2016) | Female WBT (August 2016) |
|------------------------|------------------------|--------------------|-----|--------|--------|------------------------|--------------------------|
| Single scull (x)       | +++++                  | ++                 | ++  | +      | +++    | 6:30:7                 | 7:07:7                   |
| Rowing pair (-)        | +++++                  | +++                | ++  | +      | ++     | 6:08:5                 | 6:49:1                   |
| Rowing double (x)      | +++++                  | +++                | ++  | +      | ++     | 5:59:7                 | 6:37:3                   |
| Rowing four (-)        | +++++                  | ++++               | ++  | +      | ++     | 5:37:8                 | 6:14:4*                  |
| Rowing quad (x)        | +++++                  | ++++               | ++  | +      | ++     | 5:32:3                 | 6:06:8                   |
| Rowing eight (-)       | +++++                  | ++++               | +++ | +      | +      | 5:18:7                 | 5:54:2                   |
| Lightweight double (x) | ++++                   | +++                | ++  | +      | ++     | 6:05:4                 | 6:47:7                   |

--single oar; x=double oars; PS=peak speed; WBT=world best time; \*=non-Olympic

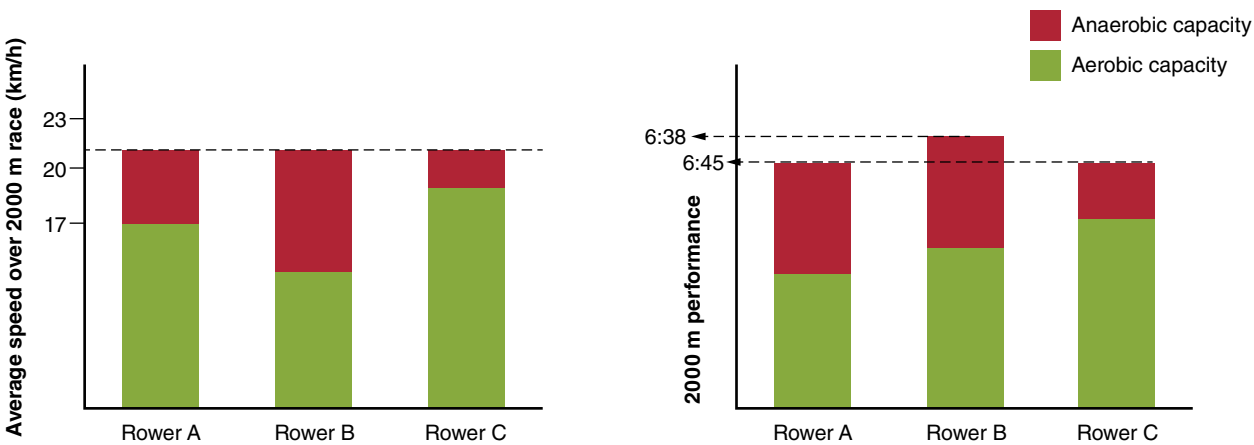
rowing event), these percent energy system contributions change slightly relative to both the boat class (table 16.1) and individual. This indicates that the relative importance of each energy pathway or physical attribute (figure 16.4) should be viewed at a boat-by-boat level. Indeed, successful World and Olympic rowing champions in the single sculls are known to spend considerable time cycling at an aerobic intensity. A boat class that has a greater aerobic contribution than other boat classes may indicate why such a strategy is successful for the single scull. Indeed, the current Olympic champion Mahe Dry-

sdale performed ~53% of his total training time cycling in the year 2012 before his gold medal success at the London Olympics. The proposed importance of each physiological requirement is outlined in table 16.2 and figure 16.5.

The energy system contributions and physical requirements not only vary on a boat-to-boat basis, but also may vary equally at an individual level. Indeed, two rowers producing the same power output over 2000 m might do so using different energy system contributions (16). However, as shown using theoretical data (figure 16.5), we should aim to



**Figure 16.5** Theoretically derived aerobic (AeR) and anaerobic (AnT) energy system contributions to the 2000 m rowing event. The percent contribution from the various energy pathways can vary in accordance with boat class. In this example, differences are apparent when comparing the M8+ (red-dotted line) and M1x (gray line).



**Figure 16.6** Theoretically differing individual energetic contributions over 2000 m (rowers A, B, and C). The aim of the training program should be to develop either (or both) systems to improve overall performance.

increase various aspects of these energy pathways to increase total power over the 2000 m event using an appropriate stimulus of training.

In summary, the physical attributes of successful rowers are predominantly aerobic endurance, with a relatively high strength component. However, as shown (figure 16.6), anaerobic glycolytic metabolism is also important. The key target of training often

is to build or maintain a rower's natural area of talent, without compromising any weakness (if any; i.e., figure 16.6). When all systems are maximized, high levels of performance may be achieved. HIIT plays a pivotal role in optimizing the required physical attributes, and when used appropriately, becomes a critical tool in the coach's and support team's toolbox.

## Key Weapons, Manipulations, and Surveillance Tools

Like other endurance sports (14, 15, 22), most rowing training hours (15 to 25 h per wk) are spent accumulating large volumes of training at low exercise intensities ( $<LT_1$ , or SR of 18 to 22 SPM). Accordingly, given the dominance of the aerobic system within the rowing event (70% to 80%), this part of the training plays an important role, as it develops the aerobic foundation needed to support the development of  $\dot{V}O_{2\max}$ , economy, and thresholds ( $\dot{V}O_2$  at blood lactate thresholds; chapter 3) (9, 19). As well as the large amounts of time training at low intensities, rowers also complete specific pace work at medium, or mid-zone, intensities. This is often known as threshold training, and the relevant intensities are shown in figure 16.2. These pieces are typically longer in duration (i.e., 5 or 6  $\times$  2 k, 2  $\times$  4 k, 2  $\times$  5 k, or 3  $\times$  3 k), with the total distance of work ranging from 8 to 12 km.

As with other endurance sports, HIIT contributes less to the total training time in rowing, but is a very important part of the overall training program design. This is particularly the case in the final build to key events, such as a World Championship or Olympic Games. Rowers generally follow the polarized model of training, with roughly 80% of training performed as low intensity ( $<LT_1$ ), and about 20% of the total training in the form of HIIT. The HIIT work is generally completed at, or just above, race intensity. HIIT is vital for developing the oxidative, anaerobic, and neuromuscular/musculoskeletal properties needed to enable fatigue resistance over prolonged durations (see chapter 3). The key weapons or HIIT formats used to achieve this, including their manipulations, will be described.

### HIIT Weapons, Targets, and Manipulations

The most common HIIT formats or weapons included in rowing training programs primarily involve long-interval, short-interval, and sprint-interval training (above 2000 m race pace). The purpose of these

intervals is to aim for type 1 through 5 targets, with a metabolic emphasis. Repeated sprint training, which incorporates very short-duration efforts (3 to 7 s), are generally used for start practices. Although physiological intensity (in terms of  $\dot{V}O_{2\max}$  and HR) is described below, in practice on water, training is typically prescribed via stroke rate and/or speed (pace per 250/500 m).

### Long Intervals

Long-interval HIIT usually takes the form of 2 to 8 min (rowing 500 to 1500 m of distance), with work intensities of 85% to 95% of HRmax ( $\geq 2000$  m race SR), 85% to 100% of  $\dot{V}O_{2\max}$ , generally using work:rest ratios between 1:0.5 and 1:2 on water. On the rowing ergometer this would mean 95% to 97% of 2000 m ergometer power. Total accumulated duration at these intensities during such interval sets is typically 20 to 40 min split between 3 and 8 repetitions, aiming generally for type 2, 3, and 4 targets. These different targets can be achieved by manipulating the work:rest ratio. For example, reducing the work period ( $<90$  s) and increasing the rest period ( $>90$  s) alters the focus of the session toward anaerobic metabolism (type 3). Conversely, increasing the work period ( $>3$  min) and reducing the recovery period ( $<3$  min) emphasizes the aerobic metabolic response (type 4; table 16.3).

### Short Intervals

Short-interval HIIT for rowing usually takes the form of 30 to 60 s work durations at intensities of 100% to 102% of race speed (normally measured in time pace per 250/500 m), generally with work:rest ratios between 1:0.5 and 1:1. For typical rowing prescription, short interval durations are typically between 30 and 60 s, or use a maximal number of strokes from a stationary position or from a rolling start (i.e., 20 to 30 maximal strokes).

Total accumulated work duration for these interval sets is generally between 10 and 20 min, split from 1 to 5 sets. Example sets can be seen in table 16.4. For elite rowers, the aim is to perform short intervals at a target stroke rate above that which they would race at, with a corresponding speed of

**Table 16.3** Typical Long-Interval HIIT Parameters for Rowing Training

| Mode   | Work intensity (% $\dot{V}O_{2\max}$ ) | Stroke rate (spm) | % of WBT (elites) | Work duration | Rest intensity | Rest duration | Rep number | Series | Target types |
|--------|----------------------------------------|-------------------|-------------------|---------------|----------------|---------------|------------|--------|--------------|
| Rowing | 85%-100%                               | 32-38             | 97%-100%          | 3 min         | 50%            | 2 min         | 4          | 1      | 3 and 4      |

**Table 16.4** Typical Short-Interval HIIT Parameters for Rowing Training

| Mode   | Work intensity (% $\dot{V}O_2\text{max}$ ) | Stroke rate (spm) | % of WBT (elites) | Work duration | Rest intensity | Rest duration | Rep number | Series | Between series recovery | Target types |
|--------|--------------------------------------------|-------------------|-------------------|---------------|----------------|---------------|------------|--------|-------------------------|--------------|
| Rowing | 100%-110%                                  | 35-42             | 100%-102%         | 30 s          | 50%            | 30            | 8          | 2      | 4 min                   | 1, 2, 3, 4   |

~100% of WBT. The desired metabolic load for these HIIT sessions includes type 1, 2, 3, and 4 targets. Similar to long-interval HIIT, these various targets can be achieved by manipulating the work:rest ratio. Alternatively, the intensity of both the work or rest period can be adjusted to shift the balance between anaerobic and aerobic metabolism.

### *Sprint Interval Training*

SIT is used frequently by rowers as competition approaches, often combined as start practice. However, if SIT is performed as maximal speed efforts (from rolling rather than standing, for example), it is important to carry out the interval with correct technique and within the functional stroke rate range. Rather than over a specific duration, SIT in rowing is normally composed of maximal efforts across a number of counted strokes (5 to 15, lasting 10 to 25 s), with recovery periods that range between 2 and 10 min. The goal of these intervals is to produce maximal force and high speed rates as fast as possible. The number of sprints is typically low (1 to 5), with time between sets relatively long (>30 min). SIT aims for type 5 targets.

### Manipulations of Interval Training Variables

The beauty of rowing is that athletes and coaches are acutely aware of the precise physiological requirements of their race. Notwithstanding unforeseen changes in weather conditions (i.e., high wind suddenly picking up to impact front, rear, or side trajectory, including wake/wave effects), each race is almost identical, taking place over a 2000 m course and typically in a relatively calm lake. As such, HIIT interval sessions can be tailored and periodized appropriately for the specific energy system requirements of a typical 2000 m race.

As mentioned, the physiological parameters of rowing include an important contribution from both anaerobic and aerobic demands, with a higher emphasis on the aerobic requirements. Accordingly,  $\dot{V}O_2\text{max}$ , power/speed associated with  $\dot{V}O_2$ , and

maximal force over short time frames (10 to 60 s) are important considerations. At the same time, however, the importance of economy, efficiency, and threshold levels cannot be overlooked.  $\dot{V}O_2$  at lactate threshold and power at onset of blood lactate accumulation (OBLA) are also important determinants of rowing success (5). These key requirements should help guide training program design and HIIT implementation in a rowing program.

The primary factors of any HIIT sessions include the intensity and duration of the work and rest periods. Typically, as most HIIT work intervals are relatively short in terms of duration, stroke rate and speed are best used to prescribe the intensity in rowing (figure 16.2). The selection of an HIIT session should be mainly governed by the appropriate physiological adaptations sought after for the individual or crew. Similarly, the number of intervals in a series, the series duration, and time between each series of HIIT intervals will depend on the specific physiologic/metabolic system being targeted (i.e., chapters 3, 4, and 5). The selection of HIIT is not always an easy task in the sport of rowing; as with crew boats, all individuals have differing physiological requirements but must perform the same training when on the water. In such instances, this sometimes means that HIIT is best performed on an indoor rowing machine so as to individualize the important HIIT session.

Manipulating the stroke rate (measure of SPM) during a work interval will impact the balance between muscular strength and coordination, as well as the neuromuscular and metabolic demands (as described in chapters 4 and 5). For example, a 3 min work interval completed at either 25 or 30 SPM will not only be performed at differing intensities but will additionally differ in the applied force and associated neuromuscular characteristics. Lower stroke rates are thought to perhaps promote development of muscular strength and alter neuromuscular demands. For long intervals, lower- and upper-end SR (e.g., 35 versus 39) (figure 16.2) provides succinctly different metabolic outputs and neuromuscular responses.

The typical scheduling of HIIT as competition approaches is also a critical consideration. Early season training generally consists of mostly aerobic training with intermittent threshold work. However, as key races approach, more focus is moved toward long and short intervals, with the final 4 wk preparatory period consisting of all three interval types (long, short, and SIT) (see tables 16.5 and 16.6).

## Surveillance and Monitoring Tools

The main surveillance and monitoring tools used in rowing include GPS devices (integrated wrist watches and on-boat devices) and heart rate. In some instances, these integrated systems can stream their data via Bluetooth and Wi-Fi to real-time systems that can be instantaneously viewed by the coach. These data can be used extensively in many applications (i.e., Training Peaks, Golden Cheetah; chapter 8) that advise the coach and practitioner regarding a rower's overall training load and training status.

## Load

Intensity prescription for rowing training will vary, but has historically been based on kilometers rowed, HR, and  $RPE \times \text{duration}$  (2). When dealing with large squads of athletes, HR monitoring can be a very attractive tool, as it has superior practical application and little practitioner intervention. However, the high-intensity and relatively short-duration HIIT intervals and the associated lag in HR response (see chapters 2, 8, and 9) prevent the use of HR for prescribing HIIT in any of the event disciplines. A more suitable approach is to monitor HIIT using rowing speed (%WBT) and time spent at targeted stroke rates (speed and SR measured via GPS and accelerometers). Rowing load can be calculated using various software systems that use Banister's classic fitness-fatigue modeling or training impulse calculation (TRIMP) scores in each respective discipline (1) (see also chapter 8).

**Table 16.5** Typical Training Week for an Elite Rower During the Preparation Period, Including a Hard and Easy Week

In the early season, typically more time is spent performing moderate-intensity training (MIT), with more HIIT applied as the race season approaches.

|           | Hard training week                                                                                                          | Easy training week                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | a.m.: 20-22 km LIT SR 20-22<br>p.m.: Rowing erg: 5×4' at HIIT, with 3' recovery (long intervals, type 4)                    | a.m.: 16-18 km LIT SR 20-22<br>p.m.: Rowing erg: 2×(10×30" at HIIT, 30" easy), 5' between repetitions (short intervals, type 3)                        |
| Tuesday   | a.m.: 24-26 km LIT SR 20-22<br>p.m.: Weight followed by 6-12 km TECH                                                        | a.m.: 24-26 km LIT SR 20-22<br>p.m.: Weights only                                                                                                      |
| Wednesday | a.m.1: 20 km row with 2×5 km MIT (SR 28-32), 5' recovery threshold<br>a.m.2: 45' LIT on indoor rowing machine<br>p.m.: REST | a.m.: 18 km row with 1×5 km MIT (SR 28-32), 5' recovery threshold<br>p.m.: REST                                                                        |
| Thursday  | a.m.: 20-22 km LIT SR 20-22<br>p.m.: Weight followed by 6-12 km TECH                                                        | a.m.: 16-18 km LIT SR 20-22<br>p.m.: Weights only                                                                                                      |
| Friday    | a.m.: 20-22 km row included 4×8' MIT with bungee for added resistance<br>p.m.: 12 km row 18-22 SR LIT                       | a.m.: 20-22 km row included 6×3' MIT (3' recovery) with bungee for added resistance (long intervals, type 4)<br>p.m.: 30' on indoor rowing erg low LIT |
| Saturday  | a.m.: 26-28 km row including 1×8 km MIT threshold<br>p.m.: REST                                                             | a.m.: 26-28 km row including 2×4 km MIT threshold<br>p.m.: REST                                                                                        |
| Sunday    | REST                                                                                                                        | REST                                                                                                                                                   |

LIT = low-intensity training: blood lactate concentration <2.5 mmol/L, heart rate <80% HRmax.

MIT = moderate-intensity training: blood lactate concentration 2.5 to 4.0 mmol/L, heart rate 80% to 90% HRmax.

HIIT = high-intensity interval training: blood lactate concentration >4.0 mmol/L, heart rate >90% HRmax.

TECH = Easy row, focusing on technical skills and drills.

**Table 16.6** Typical Training Week for an Elite Rower During the Competition Period

Presentation of the competition week assumes a normal World Cup regatta, running Friday to Sunday, which is different than the Olympic Games or World Championship events that normally run over a 1 wk period. During the hard training phase, long-interval HIIT, viewed as one of the key sessions for elite rowers, is performed every 5 d.

|           | Hard training week                                                                                                                       | Competition week                                                                                                                              |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Monday    | a.m.: 16-18 km LIT (18-22 SR)<br>p.m.: 4 × 1000 HIIT with 4' recovery (long intervals, type 4)                                           | a.m.: 18 km row include 8 × 1 km MIT with 1' recovery threshold<br>p.m.: 12 km of TECH                                                        |
| Tuesday   | a.m.: 20-22 km LIT SR 20-22<br>p.m.: Weights followed by 12 km included 8 × 10 strokes all-out from start (SIT, type 5)                  | a.m.: 24-26 km LIT SR 20-22<br>p.m.: 12 km row include 6-8 × 8-12 stroke starts all-out (SIT, type 5)                                         |
| Wednesday | a.m.: 20-22 km include 2 × (10 × 1' on 1' off) 5' between sets (short intervals, type 4)<br>p.m.: REST                                   | a.m.: 12-14 km row included 1 × 1000 and 2 × 250 HIIT (5' recovery) (long and short intervals, types 3 and 4)<br>p.m.: REST (or 8 k recovery) |
| Thursday  | a.m.1: 16-18 km LIT SR 20-22<br>a.m.2: Light weights<br>p.m.: 18-20 km row (short intervals) RST 6 × 1' HIIT with 6-8' recovery (type 4) | a.m.: 12 km row include 7 strokes MIT-HIIT, 10 strokes LIT<br>p.m.: 4 km recovery                                                             |
| Friday    | a.m.: 18-20 km LIT 20-22<br>p.m.: 12 km row TECH                                                                                         | RACE DAY 1<br>(heat and quarter finals)                                                                                                       |
| Saturday  | a.m.1: 2 × 1000 (3' recovery), 2 × 750 (2' recovery), 2 × 500 HIIT (1.5') recovery) (long intervals, type 3)<br>a.m.2: 12 km recovery    | RACE DAY 2<br>(semifinals)                                                                                                                    |
| Sunday    | REST                                                                                                                                     | RACE DAY 3<br>(finals)                                                                                                                        |

LIT = low-intensity training: blood lactate concentration <2.5 mmol/L, heart rate <80% HR<sub>max</sub>.

MIT = moderate-intensity training: blood lactate concentration 2.5-4.0 mmol/L, heart rate 80%-90% HR<sub>max</sub>.

HIIT = high-intensity interval training: blood lactate concentration >4.0 mmol/L, heart rate >90% HR<sub>max</sub>.

TECH = Easy row, focusing on technical skills and drills.

### Training Status

As mentioned, the selection of an HIIT session is not an easy one to make in the sport of rowing. As with crew boats (table 16.1), all individuals will have different physiological profiles, strengths, and weaknesses. For true individualization of HIIT to take place, sessions must either be completed in the single scull or on the indoor rowing machine. In such situations, the power continuum (whereby an athlete performs time trials on the indoor rowing machine over 10 s, 60 s, 2000 m, 6000 m, and 60 min across a 1 wk period) can be a useful test for identifying the individual characteristics of rowers so as to allow them to perform an HIIT session at an individual level. While knowledge pertaining to optimizing HIIT for individual athlete profiles is lacking, it is generally believed that the aerobic system is more malleable than the anaerobic system,

so aerobic-based HIIT intervals (types 1 and 2) generally tend to be the focus in the program.

As with other endurance-based sports, rowing coaches more often than not believe that more training is usually better, so the risk of developing overtraining is high (see chapter 7). Markers of training load include those described in the cycling (chapter 15) and triathlon (chapter 19) chapters, verbal and written training diary comments, alongside more advanced tools, including morning resting heart rate variability (17) (HRV; chapter 9).

### Strategies for Structuring the Training Program Using HIIT

The typical scheduling of HIIT as competition approaches is also a critical consideration in the program (table 16.5). Early season training generally

consists of mostly aerobic training with intermittent threshold work. However, as the races approach, focus tends to shift toward long and short intervals, with the final 4 wk preparatory period consisting of all three HIIT weapons (long, short, and SIT).

Within the sport of rowing (4, 20), it is generally considered best practice to separate HIIT days by ~48 h (see chapter 9). This tends to allow for appropriate recovery of the autonomic nervous system (ANS) and means each session can be completed optimally. In a recent study by Holt et al. (4), it was shown that threshold training resulted in the greatest reductions in ANS status, as assessed by HRV. This was followed by long and SIT intervals. Conversely, performance (30 s Wingate on the rowing ergometer) was most affected (i.e., time taken to return to baseline) by long intervals (followed by threshold work and SIT, respectively). This suggests that long intervals should perhaps be avoided during a taper period close to competition, while continuous days of threshold work may lead to excessive fatigue and should therefore be appropriately placed within the program.

## Controllable and Uncontrollable Factors

Rowers will compete on several occasions per year. However, for elite rowers, the domestic season (typically three or four regattas) is purely a preparatory phase for the international rowing season. The international season (FISA World Rowing Cups) consists of three World Cups and a World Championship event. During each regatta, crews will compete in two to four races (heats, quarterfinals, semifinals, and a final). The number of races performed in depends on the number of entries. The World Cup events normally take place from May to late July, with the World Championship occurring in late August or early September.

Unlike several other elite sports (i.e., cycling, soccer, basketball, ice hockey), where competition frequency is high, the lower frequency of competition (if appropriately planned) enables sufficient recovery to enable the scheduling of HIIT. Smaller domestic rowing regattas are however used as a training stimulus leading into major events, to gain experience in more technical areas such as starts, pacing, race tactics, and competition environment experience.

Travel demands add another challenge to the training program that must be considered. International

regattas are generally performed in Europe and are reasonably close in terms of successive proximity to one another. Thus, recovery for and from travel (i.e., rest periods before and after travel) as well as time for circadian adaptation to the time zone need to be considered (18).

## Program Structure and Progression

There are several different approaches to the planning and preparation of rowing training that can be made. Generally the training planning models in rowing follow traditional theories such as classic periodization models, block periodization models, and numerous other formats (6). As the competition program is to some degree similar every year, the training program should focus on individual and crew needs, aiming at training the primary determinants of performance physiology (chapter 3; figures 16.5 and 16.6, table 16.2) needed for the key event of the year (World Championship or Olympic Games). This means that the training programs of rowers are highly unique and specific to each boat. Nevertheless, some general observations can be made across the macro-, meso-, and microcycles observed in most rowing training programs.

### Macrocycle

A rower's seasons are typically periodized around a small number of key racing objectives, requiring three or four periods of maximal performance. As mentioned, for international rowers, the key events are the FISA World Rowing Cups. Every year, there are three FISA World Rowing Cups, which take place over the period from May to late July. The championship race (Olympics or World Championship) takes place around 4 to 5 wk after the final World Cup event of the season. Domestic regattas generally take place earlier in the year, and while part of the training process, are seldom a major goal for most elite rowers. These events create the general structure of the training program from a macrocycle perspective.

### Mesocycle

A typical basic or preparatory mesocycle in rowing follows the traditional 4 wk approach, with three loading weeks and one recovery week. However, this will vary depending on the available time between races, and can include precompetition or transition mesocycles (~2 to 4 wk), competition mesocycles (~3 to 6 wk), and recovery mesocycles (~1 to 4 wk).

### Microcycle

The microcycles of a rower's training program can be variable, but mostly occur over a 7 d weekly period. When working within crews and groups of people, 7 d microcycles allow for routine and therefore make planning for a group relatively easy. Most microcycles in rowing consist of one full day of rest per week.

### Periodized Nutrition

As outlined in chapter 4, periodized nutrition can also play an important role in the training program of some rowers. Nutrition is important not only for controlling a rower's weight (particularly for lightweight rowers; see also chapter 7), but also for affecting the adaptations achieved from a training session

(14). Specifically, manipulating carbohydrate consumption and either performing sessions after an overnight fast or performing multiple sessions in a day to force the rower to train with low-glycogen stores (popularly termed *training low*; chapter 4). However, in my experience, open-class rowers, who typically consume large amounts of food that are mainly carbohydrate based, seldom achieve this. Indeed, most rowers, but not all, generally fuel for every session (before, during, and after) and follow a high-carbohydrate diet. Training fasted in the morning is also seldom carried out in the present day elite rowing environment.

### Sample Training Programs

Training programs for preparation and competition phases are shown in tables 16.5 and 16.6.

## CASE STUDY

For the 2014 World Rowing Championships held in Amsterdam, I was fortunate to work closely alongside a top coach in the New Zealand Olympic program. The coach was coaching an athlete who was racing in the female single scull (W1x). Together, we formulated a plan that we believed could help her be successful in her event.

After the initial aerobic training preparatory period, beginning 11 mo before the World Championships, training progressed to include threshold training twice per week, including once per week of longer intervals, involving a specific race target stroke rate (100% of WBT). These intervals were progressed during the season, with one of the threshold sessions replaced with repeated sprint intervals as the race season approached (long intervals remained). HIIT was kept to a minimum during the first part of the competition season overseas, with the priority aim in this period being the maintenance of health, fitness, and recovery from racing.

We knew the final 4 wk of training prior to the last 10 taper days before the World Championships would be critical. This period consisted of sprint (5 × 60 s with 6 min recovery), long (e.g., 4 × 3.5 min on, 4 min recovery), and short intervals (e.g., 10 × 1 min on, 1 min recovery) during most weeks. The long intervals (considered most important for performance for this rower) were rolled through every 5 days. The rolling through of long intervals every 5 d was a particularly successful programming strategy. However, the training content between was critical to allow each long interval session to be performed optimally. We learned this the first time the athlete attempted to perform a second long-interval session just 5 d later (sub-optimally). We found that the best recipe was (if the first long interval day was day 1) day 2 would be recovery, day 3 would be high intensity (either sprint or short intervals rotating each time), day 4 would be all aerobic low-intensity training, and then day 5 would be back to long intervals. This was a great mix. With five being an odd number, the programming puzzle had to be continuously adjusted, but this also meant the program was very dynamic and interesting for the athlete. Full rest days would be included somewhere within the 10 d mesocycles when it felt appropriate. This was often determined by the training responses, subjective metrics, and heart rate variability scores.

As a team, we believed in our process. The athlete responded by getting progressively better in nearly every session, and we were elated when she ultimately went on to win her first world title in the single scull.

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